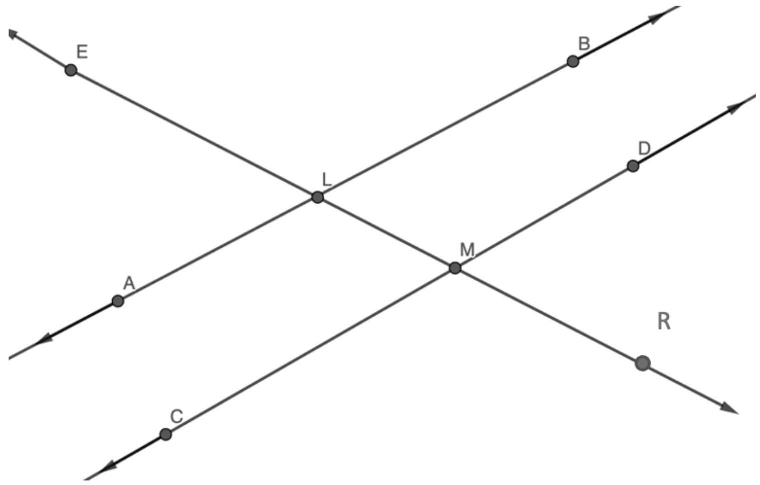
Geometry
Unit 3
Lesson 5 Homework

Name__Answer Key________
Date____________________
Period___________________

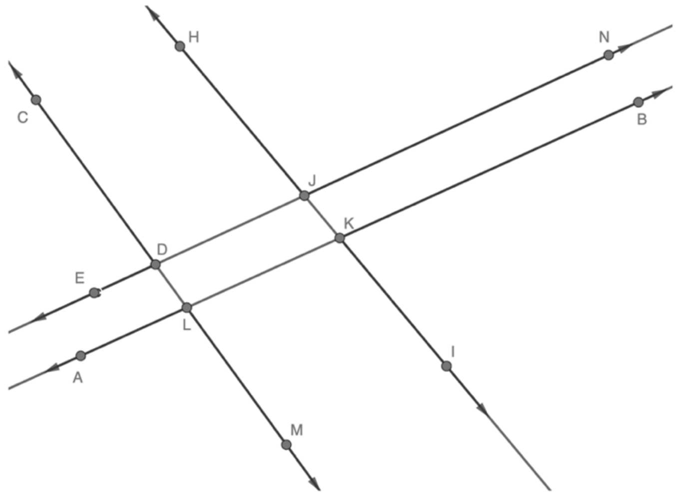
1. Given $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$, and $\overleftrightarrow{ER}$. $\overleftrightarrow{ER}$ intersects $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$.



Determine if the given condition could be used to justify that $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$?	Justification
$\angle ELA \cong \angle DMR$	✖ Yes o No	Converse alternate exterior angles
$\angle ALM \cong \angle BLE$	o Yes ✖ No	Vertical angles do not prove lines parallel
$m\angle CMR + m\angle RMD = 180°$	o Yes ✖ No	Linear pair does not prove lines parallel
$\angle ELB \cong \angle LMD$	✖ Yes o No	Corresponding angles
$\angle ELB \cong \angle RMD$	o Yes ✖ No	Congruent consecutive exterior angles do not always prove parallel lines
If $m\angle ELB = 102°$, then $m\angle ELA = 78°$	o Yes ✖ No	Linear pair does not prove lines parallel
If $m\angle RMC = 91°$, then $m\angle ALM = 89°$	✖ Yes o No	Corresponding angles are congruent in parallel lines

2. $\overleftrightarrow{CM}$, $\overleftrightarrow{HI}$, $\overleftrightarrow{EN}$, and $\overleftrightarrow{AB}$ are shown where $m\angle LKI = 94°$.



Determine if the given conditions could be used to justify that $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$. Then justify your answer.

Condition	Is $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$?	Justification
$\angle CDJ \cong \angle KLM$	o Yes ✖ No	If two lines intersected by a transversal where consecutive exterior angles are congruent, does not mean the lines are parallel.
$\angle CDE \cong \angle MLA$	o Yes ✖ No	If two lines intersected by a transversal where consecutive exterior angles are congruent, does not mean the lines are parallel.
$m\angle EDL + m\angle NJK = 180°$	✖ Yes o No	If two lines intersected by a transversal where consecutive exterior angles are supplementary, then the lines are parallel.
$\angle KLM \cong \angle HJN$	o Yes ✖ No	The two angles do not share two lines and a transversal.
$\angle JKL \cong \angle BKI$	o Yes ✖ No	Congruent vertical angles do not prove parallel lines.
If $\angle BKI = 86°$, then $m\angle BKI + m\angle LKI = 180°$	o Yes ✖ No	Angles in a linear pair do not prove parallel lines.